

1. **RS1971** John has a glass of orange juice that is $\frac{5}{8}$ full. He pours all of it into another glass and only ends up filling it halfway. The second glass has a capacity of 40 fluid ounces. What is the total capacity of the first glass in fluid ounces?

Written by: Rohan Singh

Answer: 32.

Since the second glass has a capacity of 40 fl. oz., and it is halfway full, the volume of orange juice must be 20 fl. oz.. This is the same amount that filled the first glass $\frac{5}{8}$ full. If V is the volume of the first glass, then $\frac{5}{8} \cdot V = 20$, so $V = 32$.

2. **KV1883** What is the largest number of sides a regular polygon could have such that it can be inscribed in a square?

Written by: Keshav Venkatesh

Answer: 8.

We can see that no side of the square can contain more than two vertices of the polygon, so the polygon can have at most 8 sides. When we draw what a regular octagon should look like inscribed in a square, we can see that the four sides of the squares obviously have equal length. Therefore, our answer is 8.

3. **AP2016** Lucy has a bag of marbles containing red, blue, and green marbles. The ratio of red to blue marbles is 3 : 2, and the ratio of blue to green marbles is 4 : 5. If Lucy has 45 marbles in total, how many green marbles does she have?

Written by: Aratrika Pal

Answer: 15.

For every 5 green marbles, there are 4 blue marbles and 6 red marbles. The total amount of marbles is 15. However, Lucy has 45 marbles in total, so we multiply all the values by 3, to get 15 green marbles total.

4. **JK1954** Triangle ABC is a right triangle with $AB = 2$ and $\angle BCA = 30^\circ$. Find the product of the side lengths of triangle ABC .

Written by: Joshua Koo

Answer: $16\sqrt{3}$.

It should be obvious that triangle ABC is a $30 - 60 - 90$ right triangle. Since $\angle BCA = 30^\circ$, AB (the opposite side to $\angle BCA$) must be the smallest side length of the triangle. It quickly follows that the two other sides of the triangle have length $2\sqrt{3}$ and 4. The product of these three side lengths is $16\sqrt{3}$.

5. **AZ1781** Jax has 10 pyramids, 2 cubes, 4 octahedrons, 12 cylinders, and 15 spheres. He places all of these objects in a drawer and draws an object randomly. If a is the minimum number of draws to guarantee that Jax has all of at least 1 type of object, and b is the minimum number of draws to guarantee that he has all 12 cylinders, what is $\frac{a}{b}$?

Written by: Alexander Zhao

Answer: $\boxed{\frac{39}{43}}$

First, if there are n of the same type of object, it's possible for Jax to draw $n - 1$ of these objects before he will have all of them. So, the minimum amount of draws would be $(10 - 1) + (2 - 1) + (4 - 1) + (12 - 1) + (15 - 1) + 1 = 39$. The minimum amount of draws to guarantee to have all the cylinders is $10 + 2 + 4 + 12 + 15 = 43$.

The answer is $\boxed{\frac{39}{43}}$.

6. **AP2019** A rectangular garden is 15 feet long and 10 feet wide. If a path of uniform width is built around the garden on all four sides, increasing the total area to 300 square feet, what is the width of the path as a decimal in feet?

Written by: Aratrika Pal

Answer: $\boxed{2.5}$

Let the width of the path be x feet. The original area of the garden is $15 \times 10 = 150$ square feet. When the path is added, the new length of the garden including the path becomes $15 + 2x$ feet, and the new width becomes $10 + 2x$ feet. The new area of the garden including the path is $(15 + 2x)(10 + 2x) = 300$. Expanding this equation: $150 + 20x + 30x + 4x^2 = 300$. Simplifying: $4x^2 + 50x - 150 = 0$. Solving for x , we get $x = 2.5$ or $x = -15$. We can disregard the second solution, since distance must be positive. The width of the path, therefore, is 2.5 feet.

7. **KV1991** How many regular polygons have an integer value for their angle measure?

Written by: Keshav Venkatesh

Answer: $\boxed{22}$

If the interior angle measure of a regular polygon is an integer, then its exterior angle measure is also an integer. Since the exterior angle measure of an n -gon is $\frac{360}{n}$, we have $\frac{360}{n} = k$, where k is the exterior angle measure. Since $0 < k < 180$, all the factors of 360 besides 1 and 2 are valid values for n . Therefore, our answer is $\boxed{22}$.

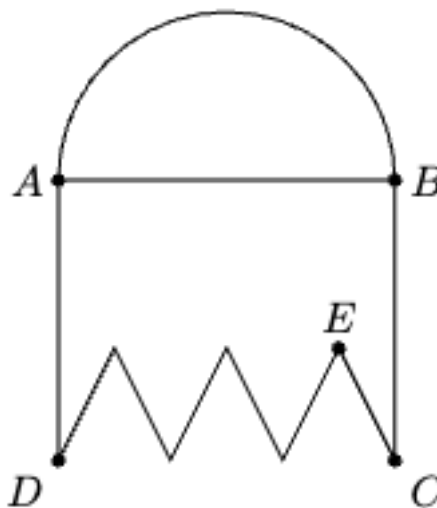
8. **NA1978** Find the units digit of $2^{2024} \times 3^{2024^{2024}} \times 7^{2024^{2024^{2024}}}$.

Written by: Nicholas Andrison

Answer: 6.

The last digit of the powers of 2, 3, and 7 all repeat in a cycle that depends on the remainder of the power when divided by 4. Since 2024 is divisible by 4, it will always have a remainder of 0 no matter what its power is. Using the pattern cycles, we can conclude that the last digit of 2^{2024} is 6, the last digit of 3^{2024} is 1, and the last digit of 7^{2024} is 1. Multiplying these numbers gives us 6, so the last digit of the original number is 6.

9. **TY1889** A Pac-Man ghost is made by joining semicircle $\widehat{AB}$ of radius 3 with rectangle $ABCD$ with 3 congruent isosceles triangles cut out, as shown below. Given that $BC = 5$ and $CE = \sqrt{5}$, find the area of the ghost in terms of π .



Written by: Troy Yang

Answer: $\frac{9\pi}{2} + 24$

Let P be the foot of the altitude from E to $\overline{CD}$. From rectangle $ABCD$, $CP \cdot 6 = 3 \cdot 2$, so $CP = 1$. By the Pythagorean Theorem, $EP = \sqrt{\sqrt{5}^2 - 1^2} = 2$. Thus, the area of each isosceles triangle is $\frac{1}{2} \cdot (2 \cdot CP) \cdot EP = 2$. The area of the ghost is then given by $\frac{1}{2} \cdot 3^2 \cdot \pi + 6 \cdot 5 - 3 \cdot 2 = \frac{9\pi}{2} + 24$.

10. **DS1780** At Mustang High, there are 3 different grade levels (10^{th} to 12^{th}) with an equal number of students in each grade. At the start of every year, around 10% of the students from each grade level receive a trophy that they keep throughout high school. Assume that trophy-winners are chosen uniformly and randomly, and note that a student may have more than one. After 5 years have passed, what percent of the entire

school population is expected to have at least one trophy? The school population does not include those that have graduated.

Written by: Dev Saxena

Answer: 18.7.

We can split this up into each grade level. First off, only 10% of the 10th graders received a trophy. The 11th graders had two different years in which they could have earned a trophy. The first year, around 10% of them received a trophy, and the second year, another 10% received one. However, we must subtract off people who have received more than one trophy. This occurs with around 1% of people (since there is a 10% chance the first year and a 10% chance the second year). Therefore, around $10\% + 10\% - 1\% = 19\%$ of the 11th graders have a trophy. Finally, we replicate the process for 12th graders. Around 19% of them have a trophy from the first two years, and another $10\% - (10\% \cdot 19\%) = 8.1\%$ received one their third year.

To summarize, 10% of the 10th, 19% of the 11th, and 27.1% of the 12th graders have a trophy. Since there are an equal number of students per grade, we average these values to get our answer, which is 18.7%.

11. **SM1882** Positive integers x and y satisfy the equation

$$\frac{15}{3+x} = \frac{12+y}{7}$$

Find the sum of all possible values of xy .

Written by: Sergei Makarevich

Answer: 30.

We get that $(x+3)(y+12) = 3 \cdot 5 \cdot 7$, since x, y , are positive integers, then we get that $(x, y) = (2, 9), (x, y) = (4, 3)$. This gives us $18 + 12 = 30$.

12. **TY1743** Jack and Jill are playing a not-so-fair game. They have a deck of 9 cards, each card numbered 1 through 9. Jack randomly takes 6 cards from the deck and gives the other 3 to Jill. Jack wins if the sum of his cards is greater than the sum of Jill's cards, and loses otherwise. How many different hands of 6 cards are winning for Jack?

Written by: Troy Yang

Answer: 82

Jack can have a total of $\binom{9}{6} = 84$ different hands. Let us count the number of hands in which Jill wins, and subtract that from 84 to find how many hands Jack wins. The sum of all cards in the deck is equal to $1 + 2 + \dots + 9 = \frac{9 \cdot 10}{2} = 45$. Note that 45 is odd, so Jack and Jill cannot tie. If Jill has a sum of x , then Jack has a sum of $45 - x$. For Jill to win, $x > 45 - x \Rightarrow x > 22.5$. Jill's biggest possible hand is (7, 8, 9), which has a sum of 24. The only other possible hand is (6, 8, 9) since lowering the value of any of

these 3 cards will result in a sum less than 22.5. Thus, Jill has 2 winning hands, and Jack has $84 - 2 = \boxed{82}$ winning hands.

13. **EL2057** Alice and Bob are blowing bubbles together. Alice can blow 2 bubbles per second for 10 seconds, then catches her breath for 2 seconds before continuing. Her bubbles last 15 seconds before popping. Bob can blow 3 bubbles per second for 9 seconds, then catches his breath for 6 seconds. His bubbles last 16 seconds before popping. If Alice and Bob start blowing bubbles at the same time, after 2 minutes, what is the positive difference between the number of Alice's bubbles and the number of Bob's bubbles still in the air?

Written by: Emma Li

Answer: $\boxed{5}$

Notice that we only need to worry about the bubbles that were blown 15 seconds before time for Alice and 16 seconds before for Bob, as any bubbles blown before that would have popped already.

For Alice, her bubbles are blown in 'cycles' of $10 + 2 = 12$ seconds, with $2 * 10 = 20$ bubbles blown per cycle. Going 15 seconds backwards, she can fit in a full cycle and the last three seconds of another. Two of those seconds will be spent resting, but she still blows 2 bubbles in the third second. Thus, she blows $20 + 2 = 22$ bubbles that will still be in the air.

For Bob, his bubbles are blown in 'cycles' of $9 + 6 = 15$ seconds, with $3 * 9 = 27$ bubbles blown per cycle. Going 16 seconds backwards, he can fit in a full cycle and the last second of another, of which will be spent resting. Thus, he blows 27 bubbles that will still be in the air.

Then, the positive difference between them is $|22 - 27| = \boxed{5}$ bubbles.

14. **TL1880** What are the last two digits of 131^{45} ?

Written by: Tianlin Liu

Answer: $\boxed{51}$

Using the binomial theorem and expanding, we obtain $(130 + 1)^{45} = 130^{45} + 45C1 * 130^{44} * 1^1 + 45C2 * 130^{43} * 1^2 + \dots + 45C2 * 130^2 * 1^{43} + 45C1 * 130^1 * 1^{44} + 1^{45}$. We see that we only need to consider the last 2 terms, since all other terms end with at least 2 zeros. The last two terms are $45C1 * 30^1 * 1^{44} + 1^{45}$, the first term ends with 50 and the second term ends with 1. Therefore, our final answer is 51.

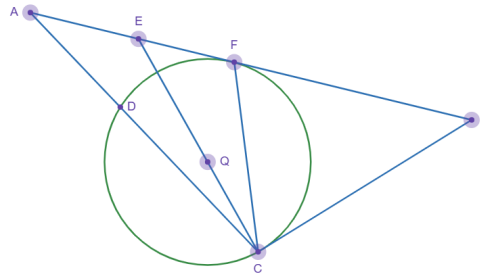
15. **TY1907** In a standard tic-tac-toe game, 5 moves have been made, i.e. there are 3 X's and 2 O's on the grid. How many board configurations exist where neither player has won yet?

Written by: Troy Yang

Answer: 1140

For the person playing X's, there are $\binom{9}{3} = 84$ ways to place the 3 X's. However, 8 of these do not work because they form 3 in a row; specifically, there are the 3 horizontal lines, 3 vertical lines, and 2 diagonals. Thus, the X-player has $84 - 8 = 76$ valid placements. The person playing O's must place 2 O's in the remaining 6 squares, giving $\binom{6}{2} = 15$ valid placements. Then, there are $76 \cdot 15 = \boxed{1140}$ total board configurations.

16. **WS1851** We have two circles O and Q tangent at point D , where the radius of O is greater than the radius of Q . Line $\overline{AC}$ is tangent to circles O and Q at points A and E , respectively. Line $\overline{BC}$ is tangent to circles O and Q at points B and F , respectively. Given that $AC = 5\sqrt{3}$ and the radius of O is 5, find the area inside quadrilateral $OACB$ but outside the two circles.



Written by: Wynn Shu

Answer: $25\sqrt{3} - \frac{100\pi}{9}$

Since $\overline{OA}$ is a radius and $\overline{AC}$ is tangent, angle $\angle OAC$ is right. By the lengths given in the problem, triangle $\triangle OAC$ is a $30 - 60 - 90$.

Let's now find the radius of circle Q . Note that $\triangle OAC \sim \triangle QEC$. Setting $OA = x$, $AC = y$, and $QE = r$, we have $\frac{x}{\sqrt{x^2 + y^2}} = \frac{r}{\sqrt{x^2 + y^2 - x - r}}$. Plugging in values gives $\frac{5}{\sqrt{25 + 75}} = \frac{r}{\sqrt{25 + 75 - 5 - r}}$. Simplifying gives $\frac{1}{2} = \frac{r}{5 - r}$ and $r = \frac{5}{3}$.

Note that triangle $\triangle OAC$ is congruent to triangle $\triangle OBC$ by SAS. Now, the area that we are looking for is the area of the two $30 - 60 - 90$ triangles subtracted by a 120° sector of circle O as well as the area of circle Q . The area of the two $30 - 60 - 90$ triangles is $2 \cdot \frac{1}{2} \cdot 5 \cdot \sqrt{3} = 25\sqrt{3}$. The circle O sector's area is $\frac{120}{360} \cdot 25\pi = \frac{25\pi}{3}$. The area

of circle Q is $\pi r^2 = \frac{25\pi}{9}$. Thus, the answer is $25\sqrt{3} - \frac{100\pi}{9}$.

17. **RRP2021** Let $f(x) = x^4 - 2x^3 + 8x^2 + x + 2$. Suppose $f(x)$ has roots r_1, r_2, r_3, r_4 . Compute

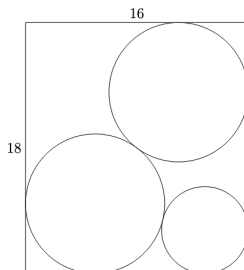
$$(r_1 + r_2 + r_3)(r_1 + r_2 + r_4)(r_1 + r_3 + r_4)(r_2 + r_3 + r_4).$$

Written by: Ryan Pascual

Answer: 2.

Note $r_1 + r_2 + r_3 = (r_1 + r_2 + r_3 + r_4) - r_4 = 2 - r_4$ by Vieta's. Then, the desired product equals $(2 - r_4)(2 - r_3)(2 - r_2)(2 - r_1) = f(2) = 36$.

18. **LT279** Refer to the diagram below. The rectangle has a width of 16 and a height of 18. Two congruent circles, ω_1 and ω_2 , are tangent to each other, and each is tangent to two sides of the rectangle. A third circle, ω_3 , is tangent to ω_1 and to two sides of the rectangle. Find the radius of ω_3 .



Written by: Linus Tang

Answer: $21 - 8\sqrt{5}$

Let the rectangle have opposite vertices $(0,0)$ and $(16,18)$ with sides parallel to the coordinate axes, and define O_1, O_2 , and O_3 to be the centers of the circles. Let R be the common radius of ω_1 and ω_2 and r be the radius of ω_3 . We can first solve for R : we know that $O_1 = (R, R)$ and $O_2 = (16 - R, 18 - R)$. Since circles ω_1 and ω_2 are tangent to each other, the distance between O_1 and O_2 is $2R$. Using the Euclidean distance formula or the Pythagorean Theorem, $(16 - 2R)^2 + (18 - 2R)^2 = (2R)^2$. This can be expanded to give a quadratic in R whose positive solutions are $R = 5$ and $R = 29$. The latter solution is obviously too large, so we have $R = 5$.

Now, we turn our attention to tangent circles ω_1 and ω_3 . We know that the distance between O_1 and O_3 is $R + r$. Furthermore, $O_1 = (R, R)$ and $O_3 = (16 - r, r)$. Again, using the Euclidean distance formula, $(16 - r - R)^2 + (r - R)^2 = (R + r)^2$. $16 - r - R = 2\sqrt{rR}$ $R + 2\sqrt{rR} + r = 16$ $(\sqrt{R} + \sqrt{r})^2 = 16$ $\sqrt{R} + \sqrt{r} = 4$ Thus, $\sqrt{r} = 4 - \sqrt{R} = 4 - \sqrt{5}$. The answer is $r = (4 - \sqrt{5})^2 = \boxed{21 - 8\sqrt{5}}$.

19. **TY1906** Consider a 13-hour analog clock, where the hour hand points to the numbers 1 through 13, evenly spread around the clock, while the minute hand still points to 60 ticks in 1 full rotation. The clock currently displays 13:00, so the minute and hour hands both point upwards. Let m be the number of minutes it takes for the minute and hour hands to first form a 144° angle, and t be the number of times the minute and hour hands form a 144° angle in the next 13 hours. Compute $m + t$.

Written by: Troy Yang

Answer: 50

To rotate 360° , the hour hand takes $13 \cdot 60 = 780$ minutes, and the minute hand takes 60 minutes. So after n minutes, the hour hand will rotate $\frac{n}{780} \cdot 360^\circ = \frac{6}{13}n^\circ$, and the minute hand will rotate $\frac{n}{60} \cdot 360^\circ = 6n^\circ$. The angle between the two hands will be $6n^\circ - \frac{6}{13}n^\circ = \frac{72}{13}n^\circ$, which is congruent to $\pm 144^\circ \pmod{360^\circ}$. Solving, we get that $n = 26 + 65k$ or $39 + 65k$, where k is an integer. m is the first positive value of n , which is 26, and each answer form of n gives $\frac{13 \cdot 60}{65} = 12$ solutions for n in a 13-hour period. Thus, $t = 2 \cdot 12 = 24$, and $m + t = 26 + 24 = \boxed{50}$.

20. **VJ1809** Max starts at the origin and can only move to the up or the right and has to take 15 or 16 total steps. How many different paths can he take, if he has to take an even number of upward steps?

Written by: Varyan Jain

Answer: $\boxed{49152}$.

If he takes 15 steps, his path must end up on a coordinate in the form of $(\text{odd}, \text{even})$ and if he takes 16 step, it must be in the form of $(\text{even}, \text{even})$. Now, let's split the cases into 15 steps and 16 steps. For 15 steps, there can be 0, 2, 4, 6, 8, 10, 12, or 14 upward steps, giving us $\binom{15}{0} + \binom{15}{4} + \binom{15}{6} + \binom{15}{8} + \binom{15}{10} + \binom{15}{12} + \binom{15}{14}$, which could be calculated but is simply half of $\sum_{n=0}^{15} \binom{15}{n}$ which can be seen if you look at $(x - y)^n$ plug in 1 for each x and y . The equation reduced down to: $\binom{n}{0} - \binom{n}{1} + \binom{n}{2} \cdots = 0$ by binomial theorem which shows that the chooses with evens and odd numbers at the bottom are equal. Since the sum of all the chooses with a particular number n at the top is 2^n , then the summation with even numbers at the bottom is 2^{n-1} . So now there's just 2^{15-1} paths for the 15-step case, and similarly there's 2^{16-1} paths for the 16-step case. Thus, $2^{14} + 2^{15} = \boxed{49152}$

21. **RT1732** A four digit number $\overline{abcd}$ is suspicious if $\overline{da} + \overline{bc} = \overline{ab} + \overline{cd}$. How many suspicious numbers are there?

Written by: Ryan Tang

Answer: $\boxed{615}$

Note that a suspicious number satisfies $10d + a + 10b + c = 10a + b + 10c + d$. Simplifying this gives $a + c = d + b$. Let this common sum be k .

For $k \leq 9$, the number of ways to choose a and c is k and the number of ways to choose b and d is $k + 1$, so the total ways is $k(k + 1) = 2\binom{k+1}{2}$. The sum of this from $k = 1$ to 9 is $2\binom{11}{3} = 330$ by hockey stick.

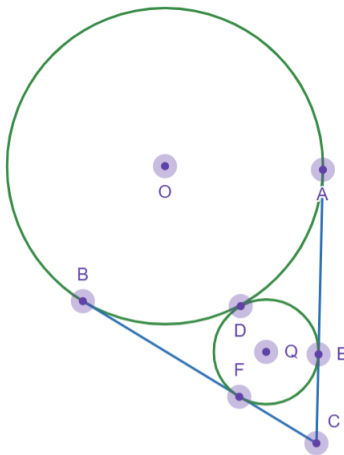
For $k \geq 10$, the number of ways to choose a and c is $9 - (k - 9) + 1 = 19 - k$ and the number of ways to choose b and d is also $19 - k$, so the total ways is $(19 - k)^2$. The sum of this from 10 to 18 is equal to $1^2 + 2^2 + \cdots + 9^2 = \frac{9 \cdot 10 \cdot 19}{6} = 285$

The sum of these numbers equals to $330 + 285 = \boxed{615}$

22. **WS1834** We have a circle Q . Lines $\overline{AB}$ and $\overline{BC}$ are tangent to circle Q at points F and C , respectively. Line $\overline{CE}$ is drawn such that it passes through the center of circle

Q . The line $\overline{AC}$ intersects circle Q at point D .

Given that $AD = \frac{16}{7}$, $AC = 7$, the area of $\triangle AEC$ is $\frac{24}{5}$, the area of $\triangle EFC$ is $\frac{24}{5}$, and $EC = 6$, find the area of triangle $\triangle BFC$.



Written by: Wynn Shu

Answer: $\boxed{\frac{96}{5}}$

Applying the power of a point theorem to point A , we have that $AF^2 = AD \cdot AC = 16$. Thus, $AF = 4$. Since triangles $\triangle AEC$ and $\triangle EFC$ have the same height and area, they also have the same base. We know that $AF = 4$, so $AE = EF = 2$. We have $A = \frac{24}{5} = \frac{1}{2}bh$, so $h = \frac{24}{5}$. Note that this is the same as the height of triangle $\triangle BFC$,

Now, to find BF , we'll first find the length of BE . Note that since QC is a radius and BC is tangent, angle $\angle ECB$ is right. By the Pythagorean Theorem, we have $EC^2 + BC^2 = EB^2$. Let's set $BC = x$. By the two-tangent theorem, we know that FB is also equal to x . It is also given that $EC = 6$, so we have $36 + x^2 = (2 + x)^2$. Solving, we have $x = 8$. Thus, the area of triangle $\triangle BFC$ is $\frac{1}{2}bh = \frac{1}{2} \cdot 8 \cdot \frac{24}{5} = \frac{96}{5}$.

23. **NA2009** Let a_1, a_2, a_3, a_4 , and a_5 be the (not necessarily real) solutions to $x^5 + 2x^4 + 3x^3 + 4x^2 + 5x + 6 = 0$. Find the value of $(a_1)^7 + (a_2)^7 + (a_3)^7 + (a_4)^7 + (a_5)^7$.

Written by: Nicholas Andrison

Answer: $\boxed{-44}$

Let $S = x^5 + 2x^4 + 3x^3 + 4x^2 + 5x + 6 = 0$. Multiplying by x on both sides gives us $xS = x^6 + 2x^5 + 3x^4 + 4x^3 + 5x^2 + 6x$. Using this, we can get $xS - S = x^6 + x^5 + x^4 + x^3 + x^2 + x - 6 = \frac{x(x^6-1)}{x-1} - 6$. We can further simplify this into $S(x-1) = \frac{x^7-7x+6}{x-1}$ and find that $S = \frac{x^7-7x+6}{(x-1)^2}$, which must equal 0. We can exclude $x = 1$ and multiply by $(x-1)^2$ to get $x^7 - 7x + 6 = 0$ or $x^7 = 7x - 6$. Thus, the roots of the original equation also satisfy this new equation, meaning $(a_1)^7 + \dots + (a_5)^7 = 7a_1 - 6 + \dots + 7a_5 - 6 =$

$7(a_1 + a_2 + \cdots + a_5) - 30$. Using Vieta's formula, $a_1 + a_2 + \cdots + a_5 = \frac{-2}{1} = -2$, and the final answer is $7(-2) - 30 = -14 - 30 = \boxed{-44}$.

24. **AZ1784** How many 6 digit numbers can be created using the digits 1, 3, 5, 6 if:

1. a prime digit cannot go after a composite digit
2. the first digit must be 1
3. every 6 must have a 1 before it.

Written by: Alex Zhan

Answer: $\boxed{337}$

IMPORTANT EDIT: Answer should be 337, below solution is incorrect, correct solution will be added sometime hopefully.

Let $f(n)$ denote the number of ways to create a n -digit number satisfying the conditions.

Now we try to find a recurrence relation to get $f(n)$.

If the last digit is 3 or 6, the previous digit cannot be 6 which yields $3 \times f(n-2)$ ways for each of these two cases.

If the last digit is 6, the previous digit must be 1 which yields $f(n-2)$ cases.

If the last digit is 1, then there are $f(n-1)$ ways.

Thus, $f(n) = 7 \times f(n-2) + f(n-1)$.

Since the first digit is 1, $f(1) = 1$.

By simple casework, we can get that $f(2) = 4$. Now, we use our recurrence relation. $f(3) = 11, f(4) = 39, f(5) = 116, f(6) = 389$.

Thus, our answer is 389.

25. **LT280** Let $S = \{1, 3, 7, \dots, 993, 997, 999\}$ be the set of positive integers less than 1000 that are divisible by neither 2 nor 5. The product of the elements of S is computed. What is the remainder when this product is divided by 1000?

Written by: Linus Tang

Answer: $\boxed{1}$

First, note that the elements of S are the positive integers less than 1000 that have a units digit of 1, 3, 7, or 9.

Let $P = 1 \cdot 3 \cdot 7 \cdot \dots \cdot 993 \cdot 997 \cdot 999$. We want the remainder when P is divided

by 1000. One way to solve the problem is to put the elements in groups of 4: $P = (1 \cdot 3 \cdot 7 \cdot 9)(11 \cdot 13 \cdot 17 \cdot 19) \cdots (991 \cdot 993 \cdot 997 \cdot 999)$.

However, there is a better grouping that allows the problem to be solved with easier computations. First, replace 997 and 999 with -3 and -1 , as this does not affect the value of the product modulo 1000: $P \equiv (-3) \cdot (-1) \cdot 1 \cdot 3 \cdot 7 \cdots 991 \cdot 993$

Now, put the elements in groups of 4: $P \equiv ((-3) \cdot (-1) \cdot 1 \cdot 3)(7 \cdot 9 \cdot 11 \cdot 13) \cdots (987 \cdot 989 \cdot 991 \cdot 993)$

Each group is of the form $p_a = (10a - 3) \cdot (10a - 1) \cdot (10a + 1) \cdot (10a + 3)$ for some value $a \in \{0, 1, \dots, 99\}$. We have $(10a - 1)(10a + 1) = 100a^2 - 1$ and $(10a - 3)(10a + 3) = 100a^2 - 9$. Now, $p_a = (10a - 3) \cdot (10a - 1) \cdot (10a + 1) \cdot (10a + 3) = (100a^2 - 1)(100a^2 - 9) = 10000a^4 - 1000a^2 + 9 \equiv 9 \pmod{1000}$.

So, working modulo 1000, we have $P \equiv p_0 p_1 p_2 \cdots p_{99} \equiv 9 \cdot 9 \cdot 9 \cdots 9 = 9^{100}$.

Now, it remains to simplify 9^{100} modulo 1000. One method is to apply the binomial theorem: $9^{100} = (10 - 1)^{100} = \binom{100}{100} \cdot 10^{100} + \cdots + \binom{100}{2} \cdot 10^2 + \binom{100}{1} \cdot 10^1 + \binom{100}{0} \cdot 10^0 = \cdots + 4950 \cdot 100 + 100 \cdot 10 + 1 \cdot 1$ (where the $\cdots$ only hides terms that are divisible by $10^3 = 1000$) $\equiv \boxed{1} \pmod{1000}$.